DTS AND COMPUTER CONTROL

1. a) To take in account the effect of the zero-order hold, we consider open-loop transfer function

$$C(s)\widetilde{H}_0(s)P(s) = \frac{k_p s + 2}{s(s-1)(\frac{s}{20}+1)}.$$

The corresponding closed-loop characteristic polynomial is

$$s^3 + 19s^2 + (20K_p - 20)s + 40.$$

Applying the Routh criterion yields that the closed-loop system is asymptotically stable for $K_p > \dfrac{21}{19} \approx 1.1053$.

[4 marks]

b) The controllers are given by

$$C_B(z) = C(s)\Big|_{s=10\frac{z-1}{z}} = \frac{21z-20}{5z-5}$$

and

$$C_T(z) = C(s)\Big|_{s=20\frac{z-1}{z+1}} = \frac{41z-39}{10z-10}.$$

Both controllers have a pole at $z = 1$.

[4 marks]

c) The equivalent discrete-time model is

$$\begin{aligned} HP(z) &= (1-z^{-1})Z\left(\frac{1}{s(s-1)}\right) \\ &= \frac{e^{0.1}-1}{z-e^{0.1}}. \end{aligned}$$

[4 marks]

d) The poles of the closed-loop system are the roots of the characteristic polynomial

$$5z^2 + (16e^{0.1} - 26)z + 20 - 15e^{0.1},$$

namely

$$z_{B1} \approx 0.9171 \qquad z_{B2} \approx 0.7464.$$

[4 marks]

e) The poles of the two controllers are very close. The reason is because the sampling time is relatively small. If the sampling time were larger, then the two discretization algorithms would yield different controllers.

[4 marks]

Answers

2. a) By the properties of the z-transform we obtain

$$(1-2z^{-1}+z^{-2})Y(z) = (2z^{-1}+z^{-2})u(k-2)$$

which reordered yields

$$\frac{Y(z)}{U(z)} = \frac{1+2z}{(z-1)^2}.$$

[2 marks]

b) The problem has a unique solution if $m = n+p-1$, where m is the order of the denominator of the controller, n is the order of the denominator of the transfer function and p is the number of performance specifications. In this case, $p = 0$, $n = 2$ and $m = 1$, so the problem has a unique solution. The characteristic polynomial is

$$\begin{aligned} p(z) &= (z+v_1)(z-1)^2 + (s_0 z + s_1)(1+2z) \\ &= z^3 + (v_1 + 2s_0 - 2)z^2 + (1 - 2v_1 + s_0 + 2s_1)z + v_1 + s_1 \end{aligned}$$

and we want that $p(z)$ satisfies

$$p(z) = z^3.$$

This yields the system of equations

$$\begin{aligned} v_1 + 2s_0 - 2 &= 0, \\ 1 - 2v_1 + s_0 + 2s_1 &= 0, \\ v_1 + s_1 &= 0, \end{aligned}$$

which has solution $v_1 = \frac{4}{9}$, $s_0 = \frac{7}{9}$, $s_1 = -\frac{4}{9}$. Thus, the controller is

$$C(z) = \frac{7z-4}{9z+4}.$$

[6 marks]

c) The closed-loop transfer function is

$$\frac{Y(z)}{R(z)} = \frac{C(z)HP(z)}{1+C(z)HP(z)} = \frac{14z^2 - z - 4}{9z^3} = \frac{14}{9}z^{-1} - \frac{1}{9}z^{-2} - \frac{4}{9}z^{-3}.$$

[3 marks]

d) The reference signal $r(k) = 1$ has z-transform

$$R(z) = \frac{1}{1-z^{-1}} = 1 + z^{-1} + z^{-2} + z^{-3} + \ldots.$$

The z-transform of the output is

$$\begin{aligned} Y(z) = \frac{C(z)HP(z)}{1+C(z)HP(z)}R(z) &= \left(\frac{14}{9}z^{-1} - \frac{1}{9}z^{-2} - \frac{4}{9}z^{-3}\right)(1 + z^{-1} + z^{-2} + z^{-3} + \ldots) \\ &= \frac{14}{9}z^{-1} + \frac{13}{9}z^{-2} + z^{-3} + z^{-4} + \ldots \end{aligned}$$

Thus, the output sequence is

$$y(0) = 0, \qquad y(1) = \frac{14}{9}, \qquad y(2) = \frac{13}{9}, \qquad y(k) = 1, \text{ for } k \geq 3.$$

[6 marks]

Answers

e) Note that the open-loop transfer function has two poles in 1. Thus, the system is of type 2. Consequently:

i) For large k, $e(k) = 0$ because there is no steady-state error for constant or ramp references.

ii) For large k, $e(k) = \alpha$ with $\alpha \neq 0$. There is a finite steady-state error for parabolic references.

iii) $e(k) \to \infty$, as $k \to \infty$.

[3 marks]

Answers

3. a) As hinted, substituting $t = (k+1)T$ and $t_0 = kT$ in

$$x(t) = e^{F(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{F(t-\tau)}Gu(\tau)d\tau.$$

yields

$$x((k+1)T) = e^{FT}x(kT) + \int_{kT}^{(k+1)T} e^{F((k+1)T-\tau)}Gu(\tau)d\tau.$$

Notice that in the interval $[kT,(k+1)T)$, $u(t)$ is identically equal to $u(kT)$. Thus, it can be brought outside the integral. Again as hinted, using the change of variable $\lambda = (k+1)T - \tau$ yields

$$x((k+1)T) = e^{FT}x(kT) + \int_{T}^{0} e^{F\lambda}G(-d\lambda)u(kT),$$

from which the relations

$$A = e^{FT}, \qquad B = \int_0^T e^{F\lambda}Gd\lambda,$$

follow. The continuous-time equation for y is:

$$y(t) = Hx(t),$$

hence, for $t = kT$ we have

$$y(kT) = Hx(kT) = Cx(kT).$$

[4 marks]

b) To compute the matrices A and B we first compute $(sI-F)^{-1}$, namely

$$(sI-F)^{-1} = \begin{bmatrix} \frac{s}{s^2+1} & \frac{1}{s^2+1} \\ \frac{-1}{s^2+1} & \frac{s}{s^2+1} \end{bmatrix}.$$

It follows that

$$A = \begin{bmatrix} \cos(T) & \sin(T) \\ -\sin(T) & \cos(T) \end{bmatrix}, \qquad B = \int_0^T \begin{bmatrix} \sin(\lambda) \\ \cos(\lambda) \end{bmatrix} d\lambda = \begin{bmatrix} 1-\cos(T) \\ \sin(T) \end{bmatrix}.$$

[4 marks]

c) The reachability matrix of the discrete-time system is

$$P = [B \quad AB] = \begin{bmatrix} 1-\cos(T) & \cos(T)-\cos^2(T)+\sin^2(T) \\ \sin(T) & -\sin(T)+2\sin(T)\cos(T) \end{bmatrix},$$

the rank of which is less than 2 if $T = k\pi$. For instance, for $T = \pi$ we have

$$P = \begin{bmatrix} 2 & -2 \\ 0 & 0 \end{bmatrix},$$

the rank of which is 1. Therefore the discrete-time system with $T = \pi$ is not reachable. For any $T \neq k\pi$, $k \in \mathbb{Z}_+$, the system is reachable.

[4 marks]

Answers

d) The observability matrix of the discrete-time system is

$$Q = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\sin(T) & \cos(T) \end{bmatrix},$$

the rank of which is less than 2 if $T = k\pi$. For instance, for $T = \pi$ we have

$$Q = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix},$$

the rank of which is 1. Therefore the discrete-time system with $T = \pi$ is not observable. For any $T \neq k\pi$, $k \in \mathbb{Z}_+$, the system is observable.

[4 marks]

e) To compute the transfer function we first determine the term

$$(zI - A)^{-1} = \begin{bmatrix} \dfrac{z - \cos(T)}{z^2 - 2\cos(T)z + 1} & \dfrac{\sin(T)}{z^2 - 2\cos(T)z + 1} \\ \dfrac{-\sin(T)}{z^2 - 2\cos(T)z + 1} & \dfrac{z - \cos(T)}{z^2 - 2\cos(T)z + 1} \end{bmatrix}.$$

The input-output transfer function is given by

$$\frac{Y(z)}{U(z)} = C(zI - A)^{-1}B = \frac{\sin(T)(z-1)}{z^2 - 2\cos(T)z + 1}.$$

For $T = k\pi$, $k \in \mathbb{Z}_+$, the transfer function is identically zero, because the system is unreachable and unobservable.

[4 marks]

Answers

4. a) The equivalent discrete-time model is

$$HP(z) = \frac{z-1}{z} Z\left(\frac{P(s)}{s}\right) = \frac{z-1}{z} Z\left(\frac{1}{s^3}\right)$$
$$= \frac{T^2(z+1)}{2(z-1)^2} = \frac{0.005(z+1)}{(z-1)^2}.$$

[4 marks]

b) The transfer function in the w-plane is (recall that $T = 0.1$)

$$HP(w) = HP(z)\Big|_{z=\left(\frac{1+0.05w}{1-0.05w}\right)} = \frac{1-0.05w}{w^2}.$$

[3 marks]

c) The acceleration constant in the w-plane is defined in the same way as in the s-plane. Thus,

$$K_a = \lim_{w\to 0} w^2\, C_1(w)\, HP(w) = K.$$

Selecting $K = 4$ yields $K_a = 4$. The open-loop transfer function is given by

$$C_1(w)HP(w) = \frac{4(1-0.05w)}{w^2}.$$

[3 marks]

d) The phase margin of the open-loop transfer function is approximately $-6°$. We need at least $56°$ more.

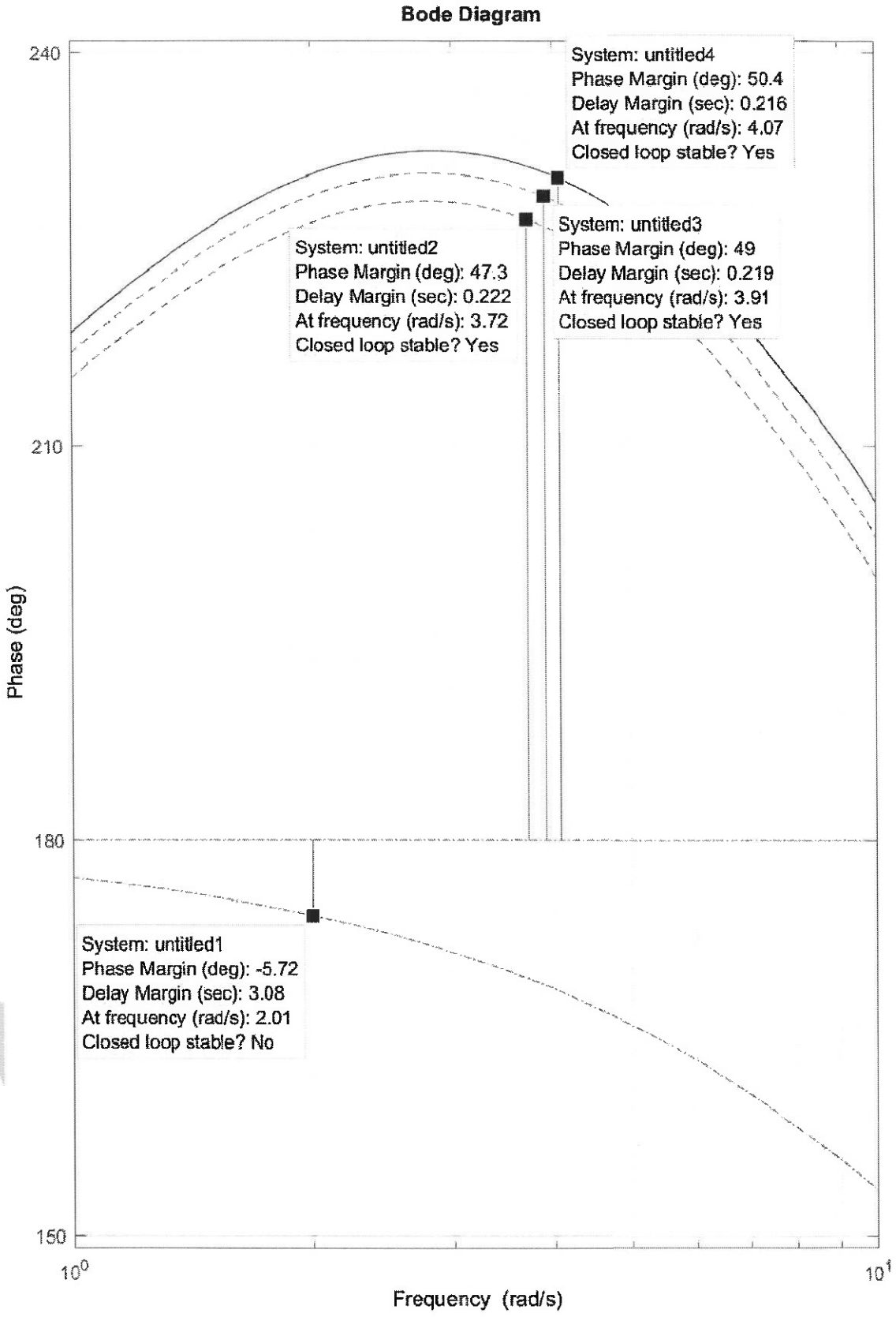
- We start selecting $m = 12$. The phase increase is approximately $58°$. The magnitude decrease is approximately 11 dB. From the Bode plot, we read that -11 dB is approximately at $\omega = 3.8$ for which the phase margin is $-11°$. Since $58 - 11 = 47 < 50°$, this compensator does not satisfy the phase margin requirement.
- We select $m = 14$. The phase increase is approximately $60°$. The magnitude decrease is approximately 11.5 dB. From the Bode plot, we read that -11.5 dB is approximately at $\omega = 3.9$ for which the phase margin is $-11°$. Since $60 - 11 = 49 < 50°$, this compensator does not satisfy the phase margin requirement.
- We select $m = 16$. The phase increase is approximately $62.5°$. The magnitude decrease is approximately -12 dB. From the Bode plot, we read that -12 dB is approximately at $\omega = 4$ for which the phase margin is $-11.5°$. Since $62.5 - 11.5 = 51 > 50°$, this is an acceptable solution.

Thus, for $m = 16$ and $\hat{\omega} = 4$, we have $\tau = \frac{\sqrt{16}}{4} = 1$ and the controller

$$C_2(w) = \frac{1+w}{1+\frac{1}{16}w}.$$

For illustrative purposes (*i.e.* this was not required) in the next page it is shown the Bode plot (phase) of the uncompensated system (dashed-dotted line), of the compensated system with $m = 12$ and $m = 14$ (dashed lines) and of the compensated system with $m = 16$ (solid line). [6 marks]

Answers


Bode Diagram
System: untitled4
Phase Margin (deg): 50.4
Delay Margin (sec): 0.216
At frequency (rad/s): 4.07
Closed loop stable? Yes
System: untitled3
Phase Margin (deg): 49
Delay Margin (sec): 0.219
At frequency (rad/s): 3.91
Closed loop stable? Yes
System: untitled2
Phase Margin (deg): 47.3
Delay Margin (sec): 0.222
At frequency (rad/s): 3.72
Closed loop stable? Yes
System: untitled1
Phase Margin (deg): -5.72
Delay Margin (sec): 3.08
At frequency (rad/s): 2.01
Closed loop stable? No
240
210
180
150
Phase (deg)
10^0
10^1
Frequency (rad/s)

e) The discrete-time controller is

$$C(z) = [C_1(w)C_2(w)]_{w=20\left(\frac{z-1}{z+1}\right)} = 37.333\frac{z-0.9048}{z-0.1111}.$$

Note that this is also equivalent to

$$C(z) = [C_1(w)]_{w=20\left(\frac{z-1}{z+1}\right)}\,[C_2(w)]_{w=20\left(\frac{z-1}{z+1}\right)}.$$

[4 marks]

Answers